

W2-Ex1 — EASY-ASSISTANT

An assistant is a name, a system prompt and a prompt template with two gaps. Every turn the backend puts the whole uploaded document into `{context}` and my message into `{user_input}`, and sends that to a local `qwen2.5:3b` through Ollama. Nothing is retrained: the model answers from the file because the file is in front of it.

THE ASSISTANT I BUILT — “TORTUGA SUPPORT”

Its document is `samples/tortuga-robotics.txt` (3,193 characters), a fact sheet for a company that does not exist — I invented it so that nothing in it could be in the model's training data, and a correct answer could only have come from the context.

System prompt

You are the support assistant for Tortuga Robotics. Answer strictly from the fact sheet you are given. Be brief and factual.

Prompt template

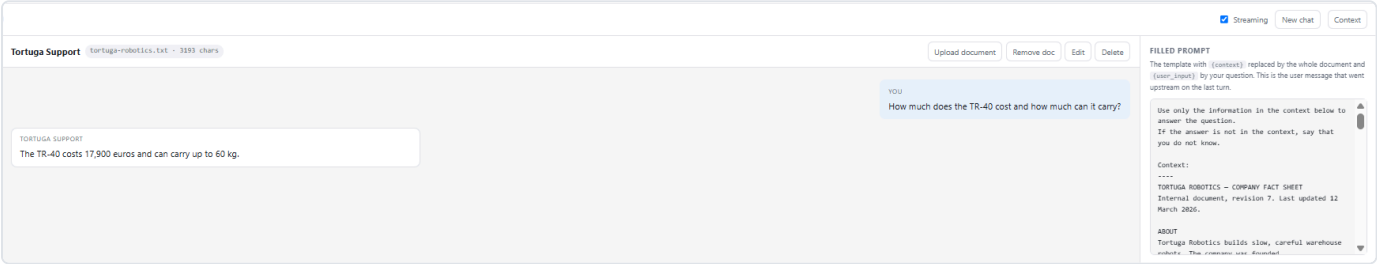
Use only the information in the context below to answer the question. If the answer is not in the context, say that you do not know.

Context:

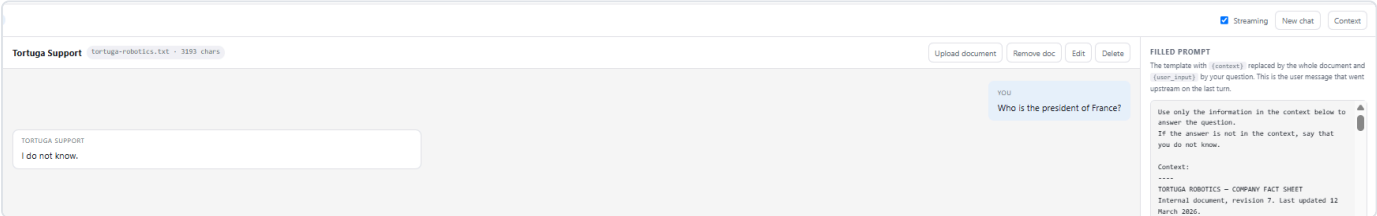
{context}

Question: {user_input}

IT ANSWERS FROM THE DOCUMENT — AND REFUSES OUTSIDE IT



Answering correctly. “How much does the TR-40 cost and how much can it carry?” → “*The TR-40 costs 17,900 euros and can carry up to 60 kg.*” Both numbers are in the fact sheet and nowhere else. The right-hand panel shows the filled prompt that was actually sent — the template with the whole fact sheet pasted into `{context}`.



Failing usefully. “Who is the president of France?” → “*I do not know.*” A model this size certainly has an opinion about France, so the refusal proves the answer is coming from my document and not from training — which is exactly the property that makes a RAG assistant trustworthy.

THE TOKEN COUNT IS THE PROBLEM

QUESTION SENT	CHARS	PROMPT_TOKENS	COMPLETION
“How much does the TR-40 cost and how much can it carry?”	54	968	25
“Who is the president of France?”	31	959	6
“And its top speed?” — asked as a follow-up	18	1,007	21

TOKENS	
Prompt	968
Completion	25
Last turn total	993
Session total	993
Watch Prompt: it does not shrink when your question gets shorter, because the whole document is sent again every turn.	

The Tokens panel after the first question.

The shortest question produced the largest prompt. My question shrank from 54 characters to 18 and `prompt_tokens` went up, from 968 to 1,007, because the 3,193-character document is re-sent in full every turn and the earlier turn had joined the history. Roughly 95% of every prompt is the same file over and over. That cannot survive a large document: a 300-page manual would be some 400,000 tokens, which overflows the context window outright — and even where it fits, I pay for the whole file on every message to use the two sentences that answer the question. Next week: retrieve only the relevant pieces.

Code: `week2/exercise1/` — the template is filled in `backend/assistants.py` (`fill_template`, one regex pass rather than `str.format()`, so a document full of braces cannot break it); the payload is built in the same file (`build_payload`); assistants and documents are stored by `backend/store.py`. Run it with `docker compose up`.